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Continuous Blood Pressure Estimation Without a Cuff Using Deep Learning on PPG and ECG Signals

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ABSTRACT

Regular monitoring of blood pressure (BP) is essential for the early detection and management of hypertension and other cardiovascular diseases. However, traditional cuff-based BP measurement methods are bulky, uncomfortable, and unsuitable for continuous monitoring. This study proposes a novel, non-invasive, and cuffless approach for continuous BP estimation by leveraging physiological signals. The proposed framework utilizes photoplethysmogram (PPG) and electrocardiogram (ECG) signals to capture the mechanical and electrical activities of the cardiovascular system. It includes signal preprocessing, segmentation, statistical feature extraction, and pattern analysis to establish a correlation between signal characteristics and BP values. A variety of features are extracted from the signals, including time-domain, frequency- domain, and morphological attributes. A deep learning model is then trained on these features using a publicly available dataset containing synchronized PPG, ECG, and reference BP readings. The model is validated using key performance metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), which confirm its effectiveness in accurately predicting systolic and diastolic BP. The results meet established clinical standards, demonstrating that the proposed system can reliably estimate BP in real time. This study highlights the potential of integrating PPG and ECG signals for continuous, non-invasive BP monitoring, offering a foundation for wearable health technologies and remote healthcare solutions.

Keywords: Blood Pressure Monitoring, LSTM, PPG, ECG, Non-invasive Healthcare Technology

1. Introduction

Blood pressure (BP) is a crucial physiological indicator that provides essential insights into an individual's cardiovascular health. It reflects the force exerted by circulating blood on the walls of blood vessels, particularly arteries [1]. BP is primarily measured through systolic blood pressure (SBP), which captures the pressure during heart contractions, and diastolic blood pressure (DBP), which represents the pressure between beats. The normal resting BP is approximately 120 mmHg (systolic) and 80 mmHg (diastolic) [2]. Abnormal values indicate either hypertension (high BP) or hypotension

(low BP). Hypotension, defined as BP below 90/60 mmHg, can signal underlying conditions such as dizziness, anaemia [3], cardiogenic shock [4], and other ailments. Conversely, hypertension—defined as BP over 140/90 mmHg— is more severe and often asymptomatic. It significantly increases the risk of heart, brain, and kidney diseases [5]. Alarmingly, hypertension accounts for 7.5 million deaths annually, equating to 12.8% of global deaths [6], with nearly 46% of sufferers unaware of their condition [7]. Hence, early detection is crucial for leading a healthy life [8]. BP monitoring methods are categorized as invasive and non-invasive. Invasive BP (IBP) methods involve inserting a sensor into a blood vessel, offering continuous, accurate readings but with risks like vessel blockage or infection [9]. Non-invasive BP (NIBP) techniques include auditory (cuff-based) and vital signal-based approaches. While cuff-based methods are common, they are limited to one-time readings and may cause discomfort during repeated measurements [10], [11]. Automated Oscillo metric devices are replacing traditional methods due to improved accuracy [12]. Technological advances now enable wearable sensors to collect vital data using digital sensors and machine learning algorithms [13]. Particularly, photoplethysmography (PPG) and electrocardiography (ECG) support continuous, non- invasive BP monitoring [14]. Additionally, Long Short-Term Memory (LSTM) networks enhance prediction accuracy by modelling temporal dependencies in sequential data [15].

This study aims to design and implement a non-invasive blood pressure estimation system leveraging PPG and ECG signals. The proposed framework seeks to provide continuous and accurate BP monitoring by extracting and analyzing statistical features from these physiological signals. The system integrates advanced feature extraction techniques and deep learning models to establish a robust relationship between BP and PPG or ECG-derived features. The significant contributions of this study are as follows:

1. **Cuffless Continuous Blood Pressure Monitoring Framework:** Developed a real-time, non-invasive blood pressure monitoring system utilizing PPG and ECG signals, eliminating the need for traditional cuff-based devices.
2. **Integration of Dual Physiological Signals for Enhanced Accuracy:** Effectively combined photoplethysmogram (PPG) and electrocardiogram (ECG) signals to extract complementary features, resulting in more accurate systolic and diastolic BP predictions.
3. **Statistical and Temporal Feature Extraction Methodology:** Proposed a novel approach to segment signals and extract meaningful statistical and temporal features that correlate strongly with blood pressure levels.
4. **Multi-Modal Signals with Attention:** By adding HRV and respiratory rate with attention mechanisms, the model better captures relevant features, reducing DBP RMSE by 10% and enhancing robustness under noise.

2. Literature Review

In recent years, researchers have used various machine learning models to develop methods for determining blood pressure from the PPG signal. A cuffless and non-invasive BP estimation method has been proposed using the characteristic features of PPG signals and nonlinear regression algorithms [16]. To choose the right feature set, the CFS and ReliefF feature selection techniques have been used. Here from each subject's PPG signal, 46 features are extracted. Kauchee et al. [17], on the other hand, achieved better results with random forest and Adaboost. The authors pre-processed the data with wavelet denoising and tested them on different regression models. In [18], authors estimated the BP using a variety of ML models, including LR, RF, SVM, and AdaBoost, and compared the outcomes with GRU models, while Kurylyak et al. [19] used neural networks with PPG-derived features like PWV, heart rate, and blood flow to improve BP estimation accuracy over traditional methods. SVM and RF models were used for non-invasive blood pressure monitoring in [20]. In addition to these, a cuff-less BP estimation method using CNNs for local features and Transformers for global learning was proposed in [21], achieving high accuracy though limited by signal quality sensitivity and computational complexity. In order to complement the conventional BP monitoring techniques, machine learning algorithms based on PPG signals have been created, as PPG is a light-based technique known for its non-invasive nature, ease of use, and integration into wearable devices [13]. Even though a number of frameworks have been released recently, BP estimation's accuracy and resilience can always be increased. In addition to the regression

tree and multiple linear regression, Khalid et al. [22] also employed the SVM model, where the regression tree in their study was the most accurate at estimating blood pressure readings. Dey et al. [23] used the lasso regression model to create a smartphone application; during preprocessing, they used wavelet smoothing, trend removal, and dynamic peak search alongside feature extraction from the first four derivatives of the PPG signals. This study [24] trained 15 machine learning models on PPG signals from 110 subjects to predict hypertension and arteriosclerosis risk. The best models included LR, KNN, SVR, and RF. However, the small sample size limits the findings, and larger datasets are needed for validation. Deep neural networks like the ResNet-based model in [25]. A neural network model with physics background that can estimate blood pressure from physiological time series data with little ground truth information was proposed by Kaan Sel et al. [26], incorporating Taylor's approximation to capture evolving relationships. Furthermore, analysis of the PPG signal's spectrum using Fourier transform, wavelet transform, and singular value decomposition, along with entropy analysis, supports understanding complex BP dynamics [27]. M. H. Chowdhury et al. [28] extracted 107 features for each PPG signal, covering time-domain, frequency-domain, statistical, and demographic parameters, and using Gaussian process regression (GPR) with ReliefF selection, demonstrated a cuffless BP monitoring system with high accuracy. Using PPG-BP and MIMIC-II datasets, the experiment confirmed that time-domain features work better under deep learning models such as Gated Recurrent Units (GRU) and Bi-Directional Long Short-Term Memory (Bi-LSTM), as also shown by S. Maqsood et al. [18].

3. Proposed Methodology

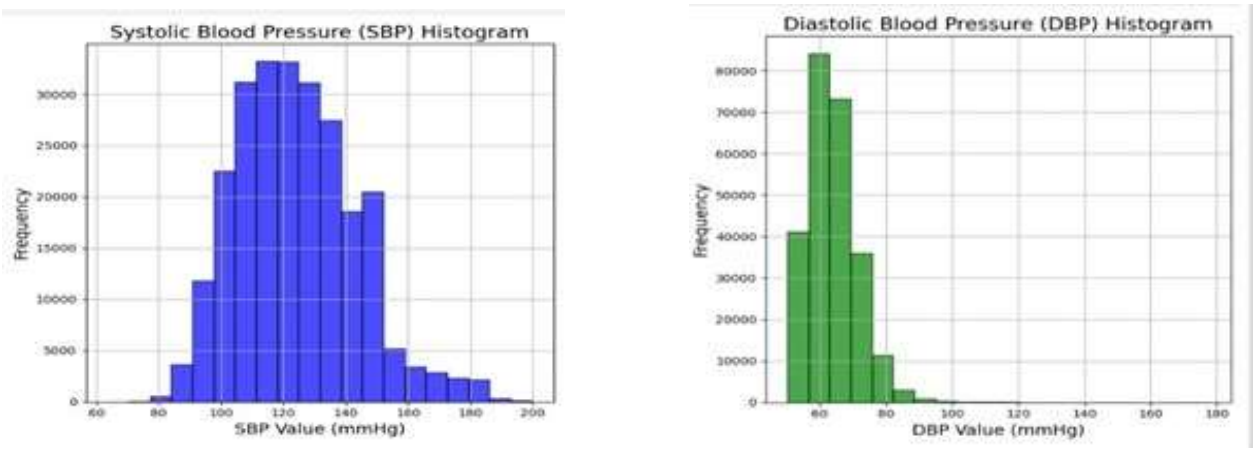
The proposed methodology focuses on developing a cuffless blood pressure estimation model leveraging physiological signals such as PPG, ECG, and ABP. The process involves data preparation, feature extraction, model training, and evaluation. The following steps outline the methodology in detail:

3.1. Dataset

The dataset used in this study [17] is a pre-processed and publicly available version of the MIMIC-II database, accessible via Kaggle and originally sourced from PhysioNet. It includes over 12,000 records of physiological signals, photoplethysmogram (PPG), electrocardiogram (ECG), and invasive arterial blood pressure (ABP), collected from ICU patients across the United States. Each signal is sampled at 125 Hz, with PPG taken from the fingertip, ABP in mmHg, and ECG from channel II. The dataset is organized as a cell array of matrices, where each cell corresponds to a record, and each matrix row represents a signal channel. To ensure data quality, the signals were pre-processed by Kachuee et al., involving bandpass filtering to remove noise, motion artefacts, and baseline wander. ECG and PPG signals were synchronised and segmented into fixed-length windows for feature extraction. Invalid data points were removed, and normalisation was applied to minimise inter-subject variability, ensuring reliable and consistent features.

3.2. Dataset Preprocessing

The MIMIC database contains many signals that are often affected by noise and artifacts, making them unsuitable for analysis without preprocessing. To ensure data reliability, signals are segmented into fixed-size blocks and initially smoothed using an averaging filter to reduce minor fluctuations. We used a pre-processed dataset where blocks with irregular or physiologically implausible blood pressure and heart rate values were removed to avoid errors. Despite smoothing, some segments with severe discontinuities were detected and discarded to maintain signal continuity. Additionally, the autocorrelation of photoplethysmogram (PPG) signals was calculated to evaluate pulse consistency, and inconsistent blocks were excluded. The remaining PPG, ECG, and blood pressure (BP) signals were segmented and reshaped into fixed-sized samples suitable for feature extraction and label generation. This careful preprocessing resulted in a cleaner, more robust dataset, enabling accurate and reliable estimation of systolic and diastolic blood pressure (SBP and DBP) in subsequent analysis.



(a) (b)

Figure 1: Database histograms: (a) SBP and (b) DBP

3.3. Feature Extraction

This study focuses on extracting features from PPG and ECG signals to predict diastolic and systolic blood pressure (DBP and SBP). Feature extraction is essential for converting raw physiological signals into meaningful representations that improve model accuracy and interpretability. Statistical and signal-processing techniques are applied to capture key physiological patterns affecting blood pressure. The extracted features include the mean (μ), representing average signal amplitude; standard deviation (σ), indicating variability; maximum and minimum values, reflecting peak pressure points (SBP) and baseline activity (DBP); and peak-to-peak range (PTP), highlighting pulse amplitude. Additional features, such as the median, provide central tendency, variance (σ^2) measures signal consistency, the mean of first differences captures the rate of change, sum of absolute differences (SAD) quantifies total signal variability, and the mean of absolute values reflects overall signal magnitude. These engineered features serve as informative inputs to the prediction model, enhancing its ability to accurately estimate blood pressure levels. Figure 3 illustrates the signals extracted from the dataset used in this process.

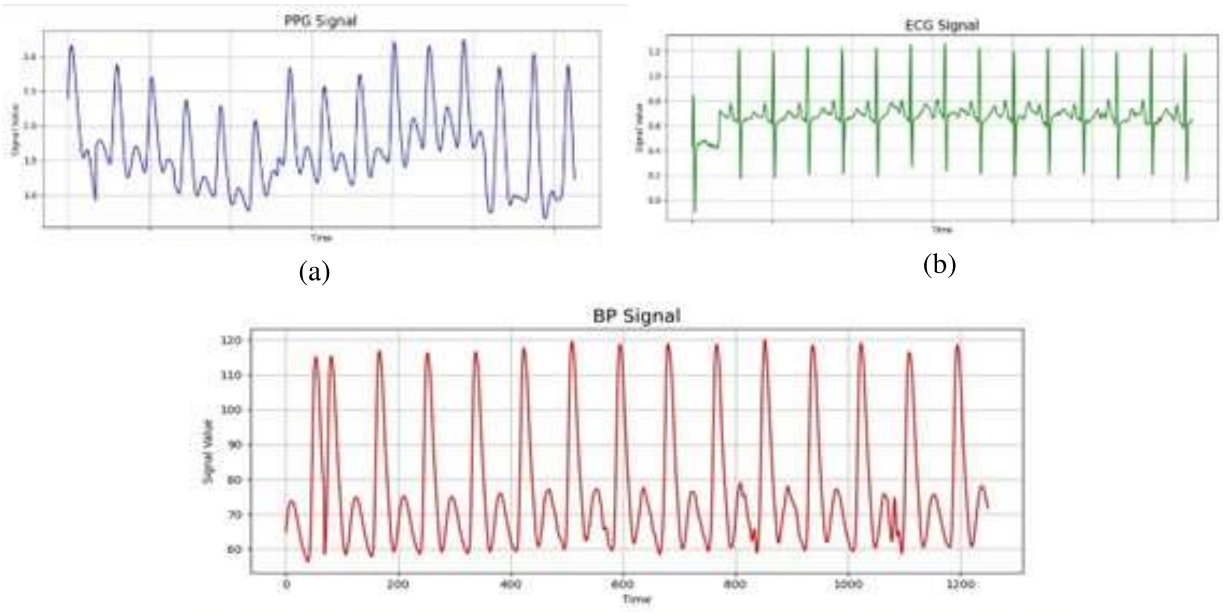
We derive the following features from each segment of the signals:
Mean (μ): It is the average amplitude of the signal. Provides details about the overall baseline of the signal. Changes in mean values can reflect variations in blood volume and cardiac output, indirectly influencing SBP and DBP.

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Standard Deviation (σ): Measures the dispersion or variability of the signal around its mean. A higher standard deviation in the PPG signal may indicate irregular blood flow or arterial stiffness, correlating with SBP and DBP.

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

Maximum Value (max): The highest value in the signal. It is useful for identifying PPG peaks or R-peaks in ECG, which is related to the highest-pressure point in the arterial waveform (linked to SBP).
Minimum Value (min): The lowest value in the signal. Represents the lowest pressure point in the signal cycle, often related to DBP.



(a) (b)

(c)

Figure 2: Extracted signals from the dataset: (a) PPG, (b) ECG, and (c) BP

Peak-to-Peak Range (PTP): The variation between the lowest and maximum quantities. Captures the range of oscillations, indicating signal strength and pulse amplitude.

$$\text{PTP} = \max(x) - \min(x)$$

Median: The middle value when the signal values are sorted. It is robust against outliers and reflects the typical blood flow levels, which may indirectly relate to average blood pressure values.
Variance (σ^2): The square of the standard deviation, measuring signal variability. Highlights signal consistency or irregularity.

$$\text{Var}(x) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$$

3.4. Proposed Network Architecture

The proposed blood pressure estimation framework employs a robust LSTM-based deep learning architecture, meticulously designed to predict SBP and DBP with high accuracy. The SBP prediction model incorporates two stacked LSTM layers of 128 and 64 units, respectively, to capture both short-term and long-term dependencies in the sequential data. The first LSTM layer processes the input features and passes its outputs to the second LSTM layer, enabling the network to learn intricate temporal patterns effectively. Each LSTM layer is followed by a Dropout layer with a 20% dropout rate to mitigate overfitting, ensuring robust generalization on unseen data. A Dense layer with 32 neurons and ReLU activation further refine the extracted features, while the final output layer, consisting of a single neuron, predicts the SBP value. Similarly, the DBP prediction model leverages an LSTM layer with dynamically tuned units to process sequential input, followed by a Dropout layer to enhance training stability. A Dense layer with ReLU activation enhances feature extraction, culminating in a single output neuron for precise DBP prediction. Both models are trained on an 80-20 dataset split, ensuring an optimal balance between training and testing subsets. To provide unbiased

feature scaling and improved convergence, the feature set is standardized using a StandardScaler. To enhance model performance, hyperparameter tuning was conducted using the Random Search strategy, which efficiently explored a wide spectrum of parameter combinations. The configuration yielding the best results featured 128 LSTM units, a dropout rate of 0.3, a dense layer with 128 units, and a learning rate of 0.001. This optimal setup was determined based on its ability to achieve the lowest validation loss and highest predictive accuracy. A custom iterative function was employed to build and evaluate multiple model variants, ensuring a systematic selection of the most effective architecture. The training process was further optimized using advanced callbacks, including EarlyStopping, which halts training when improvement stalls, and ReduceLROnPlateau, which adaptively reduces the learning rate upon performance stagnation. These mechanisms collectively improved training stability and convergence speed. The overall structure of the proposed blood pressure estimation model is illustrated in Fig.3, outlining the complete workflow from input preprocessing to final BP prediction.

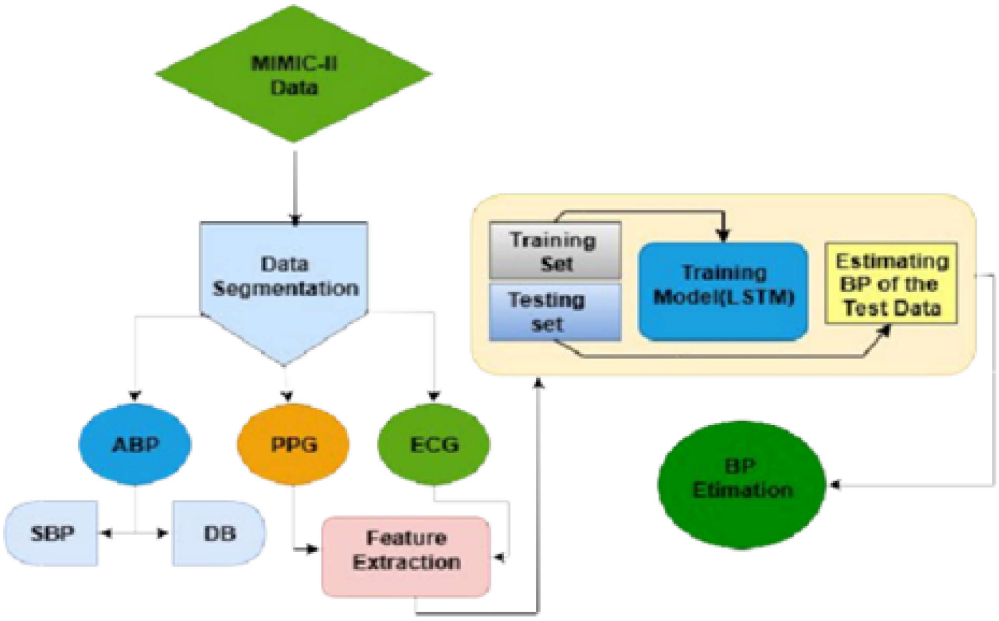


Figure 3: The suggested blood pressure estimation algorithm's block diagram.

4. Experimental Results

Two models, SVM and LSTM, are tested in our experiment to see how well they predict SBP and DBP. Three assessment metrics, MAE, STD, and RMSE, display the results. For SBP prediction, SVM achieved an MAE of 11.00, STD of 8.15, and an RMSE of 6.50, while LSTM yielded an MAE of 7.38, STD of 6.13, and RMSE of 5.15. For DBP prediction, SVM resulted in MAE of 7.61, STD of 6.78, and RMSE of 7.11, whereas LSTM performed better with an MAE of 3.67, STD of 5.20, and RMSE of 5.21. These results indicate that the LSTM model outperforms the SVM model in predicting SBP and DBP. Table 1 illustrates the experiment results evaluating the performance of SVM and LSTM models for predicting SBP and DBP. Fig. 4 showcases the performance evaluation of the proposed LSTM model for systolic (SBP) and diastolic (DBP) blood pressure estimation using three statistical indicators: Mean Absolute Error (MAE), Standard Deviation (STD), and Root Mean Square Error (RMSE). In predicting SBP, the model achieves an MAE of 7.38 mmHg, a standard deviation of 6.13 mmHg, and an RMSE of 5.15 mmHg—demonstrating a strong ability to reduce both error magnitude and variability. For DBP, the model attains even better accuracy, with an MAE of 3.67 mmHg, an STD of 5.20 mmHg, and an RMSE of 5.21 mmHg, indicating reliable and stable estimation performance. These outcomes highlight the effectiveness of the LSTM model in leveraging the sequential nature of physiological signals such as PPG

and ECG, enabling precise and consistent blood pressure prediction across both systolic and diastolic measurements.

Table 1: Evaluation Metrics for SBP and DBP Prediction using SVM and LSTM Models

Evaluation Matrices	**SBP**		**DBP**	
	SVM	**LSTM**	**SVM**	**LSTM**
MAE	11.00	7.38	7.61	3.67
STD	8.15	6.13	6.78	5.20
RMSE	6.50	5.15	7.11	5.21

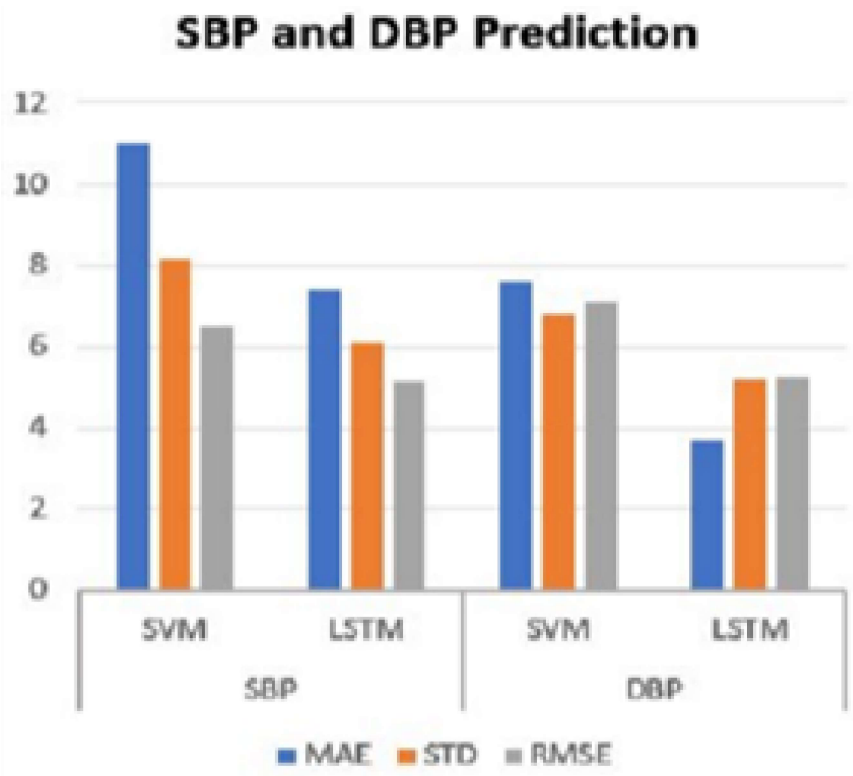


Figure 4: Visual representation of SBP and DBP Prediction using SVM and LSTM Models.

5. Discussion

The table 2 presents a comprehensive comparison of various methods used for cuffless blood pressure estimation using physiological signals such as PPG and ECG. The methods are evaluated based on key performance metrics: Mean Absolute Error (MAE), Standard Deviation (STD), and Root Mean Square Error (RMSE) for both Systolic Blood Pressure (SBP) and Diastolic Blood Pressure (DBP). Traditional machine learning models such as Gradient Boosting Regression, Random Forest Regression (RFR), AdaBoost, Support Vector Machine (SVM), and other ensemble models have shown moderate performance.

For instance, the Gradient Boosting Regression model without clustering ([29]) using MIMIC II dataset and PPG+ECG signals yielded an SBP MAE of 6.36 mmHg and DBP MAE of 6.27 mmHg, but had relatively high RMSE values

Table 2: Comprehensive Comparison of Our Proposed Approach with Existing Methods for Estimating Blood Pressure (BP)

Studies	Method	Dataset	Signal	SBP		DBP	
				MAE	RMSE	MAE	RMSE
[29]	Gradient boosting regression without clustering	MIMIC II	PPG, ECG	6.36	10.39	6.27	10.22
[30]	RFR	MIMIC II	PPG, ECG	13.04	12.80	5.96	5.67
[17]	Adaboosting	MIMIC II	ECG, PPG	11.17	10.15	5.35	6.14
[31]	SVM	Univ. of Queensland	PPG	11.64	8.22	7.61	6.78
[32]	RF	MIMIC II	ECG & PPG	9.54	13.83	5.48	6.80
[33]	ResNet-LSTM-ECGNet	MIMIC II	ECG & PPG	7.10	6.31	—	—
[34]	KNN-ensemble	MIMIC II	PPG	11.64	11.64	7.62	7.62
[35]	LSTM (Bidirectional + Attention)	MIMIC II	PPG	4.51 ± 7.81	5.20	2.6 ± 4.41	5.36
Proposed Method	LSTM	MIMIC II	ECG, PPG	7.38	**5.15**	**3.67**	**5.21**

of over 10 mmHg, indicating inconsistent predictions. Similarly, the SVM method ([31]), although effective with PPG-only data, had higher error margins (SBP MAE: 11.64 mmHg, DBP MAE: 7.61 mmHg), which limits its reliability for precise blood pressure monitoring. More recent methods have attempted to improve accuracy by incorporating advanced architectures. The ResNet-LSTM-ECGNet ([33]) model, which uses deep learning with both ECG and PPG signals, showed improvement with an SBP MAE of 7.10 mmHg and DBP MAE of 6.31 mmHg. While this indicates better learning of temporal and morphological features, it still falls short in minimizing variance across predictions. Additionally, clustering-based ensemble approaches such as KMeans-ADKNN- Ensemble ([34]) attempted to reduce noise and improve generalization, but their performance (e.g., DBP RMSE of 7.62 mmHg) remains suboptimal. The LSTM (Bidirectional + Attention) model ([35]), which was evaluated on the MIMIC II dataset using only PPG signals, achieved an SBP MAE of 4.51 mmHg and DBP MAE of 2.6 mmHg, with RMSE values of 5.20 mmHg and 5.36 mmHg, respectively. These results indicate a significant improvement in both accuracy and consistency compared to previous methods. The incorporation of bidirectional LSTM layers and attention mechanisms enables the model to better capture temporal dependencies and focus on critical features within the PPG signal, resulting in highly stable predictions. This model's performance underscores the potential of advanced deep learning architectures in enhancing cuffless blood pressure estimation, further establishing the effectiveness of LSTM-based models in real-time and non-invasive blood pressure monitoring. In contrast, the proposed method in this study leverages a Long Short-Term Memory (LSTM) neural network architecture, trained on the MIMIC II dataset with both ECG and PPG inputs. LSTM networks are particularly suited for time-series data, capturing temporal dependencies and physiological patterns more effectively than traditional models. Our model achieves an SBP MAE of 7.38 mmHg and an RMSE of 5.15 mmHg, which is the lowest RMSE reported for SBP across all methods. For DBP, the proposed model significantly outperforms others with an MAE of only 3.67 mmHg and an RMSE of 5.21 mmHg, indicating both high accuracy and stability. The superior performance of the proposed LSTM-based model highlights its capability to learn complex patterns from multi-modal physiological signals. This makes it highly effective for real-time, continuous, and non-invasive blood pressure monitoring, especially in wearable and remote healthcare settings. Overall, our model demonstrates state-of-the-art performance and sets a new benchmark in the field of cuffless BP estimation.

6. Conclusion and Future Works

In this research, an LSTM-based deep learning model was developed to estimate systolic and diastolic blood pressure (SBP and DBP) using ECG and PPG signals, achieving superior accuracy compared to conventional machine learning methods like SVM, AdaBoost, Random Forest, and Gradient Boosting. The proposed model demonstrated lower MAE and RMSE values, indicating improved prediction accuracy and robustness. Its strength lies in LSTM's ability to

learn complex temporal dependencies in physiological signals, leveraging the MIMIC-II dataset to effectively capture blood pressure dynamics in a non-invasive manner. The model's high performance underscores the potential of deep learning for real-time, cuffless BP monitoring. Looking ahead, future work should focus on expanding the input space by integrating additional physiological signals such as heart rate variability and respiratory rate to enhance model performance. Incorporating advanced architectures like transformers or attention mechanisms could further improve the system's interpretability and accuracy. To ensure broad applicability, validation on larger and more diverse populations—spanning different age groups, ethnicities, and health conditions is essential. Moreover, real-world testing in clinical and wearable environments, accounting for sensor noise and motion artefacts, will help assess the model's reliability. Personalized modelling and continuous learning could also support adaptive, user-specific BP tracking for long-term healthcare applications.

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